# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 02

**Full Syllabus (Chapters 1–11)** | Class 7 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** They may be harmful | safety |
| 2 | **(a)** sour | acids taste sour |
| 3 | **(b)** battery | combination of cells |
| 4 | **(c)** copper | a good conductor |
| 5 | **(c)** oxygen | life support |
| 6 | **(b)** Aluminium | a metal |
| 7 | **(b)** Bursting of a firecracker | explosion |
| 8 | **(c)** chemical | new substance |
| 9 | **(c)** think independently and say no | make thoughtful decisions |
| 10 | **(d)** regular bathing and clean clothes | prevents infection |
| 11 | **(a)** less dense | density decreases |
| 12 | **(c)** more than light-coloured objects | good absorbers |
| 13 | **(b)** 3 hours | time = distance ÷ speed |
| 14 | **(d)** 15 m/s | 1 km/h = 5/18 m/s |
| 15 | **(b)** small intestine | inner lining |
| 16 | **(c)** alimentary canal | digestive tract |
| 17 | **(a)** no blue-black colour | no chlorophyll |
| 18 | **(d)** stomata | on leaf surface |
| 19 | **(a)** regular reflection | clear image |
| 20 | **(a)** its position relative to the Sun and Earth | different lit portions seen |

### SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | A substance whose smell changes in acid or base (1); onion or vanilla (1). | 1+1 |
| 22 | A complete path allowing current to flow (2). | 2 |
| 23 | Burns with a bright white flame forming white magnesium oxide (2). | 2 |
| 24 | Every individual grows at their own pace (2). | 2 |
| 25 | Evaporation, condensation, precipitation (2). | 2 |
| 26 | Works at all times (2). | 2 |
| 27 | Through gills that absorb dissolved oxygen from water (2). | 2 |
| 28 | Fixes atmospheric nitrogen in root nodules (2). | 2 |

### SECTION — C (Short Answer) — Answers — 8 × 3 = 24

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 29 | Connect wire to bulb base and cell terminals and side; touch bulb base to the other terminal (3). | 3 |
| 30 | Both are chemical changes producing new substances (3). | 3 |
| 31 | Convection currents distribute heat (3). | 3 |
| 32 | Pranayama and Tummo (1.5); improve health (1.5). | 1½+1½ |
| 33 | Laterally inverted (1.5); because of left-right reversal (1.5). | 1½+1½ |
| 34 | Different hemispheres (3). | 3 |
| 35 | Formic acid is injected (1.5); base neutralises it (1.5). | 1½+1½ |
| | **OR:** Soap making and antacids (3). | 3 |

| | | |
|---|---|---|
| **36** | Make food by photosynthesis (3). | 3 |
| | **OR:** Day: photosynthesis (1); night: respiration only (1); transpiration more by day (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**4 × 5 = 20**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **37** | (i) Basic and acidic (2). (ii) Litmus test (2). (iii) MgO; $SO_2$ (1). | 2+2+1 |
| | **OR:** (i) Keep air and water away (1). (ii) Painting, oiling, galvanising (3). (iii) Saves money and resources (1). | 1+3+1 |
| **38** | (i) Blood and bones (2). (ii) Spinach, milk (1). (iii) Anaemia, weak bones (2). | 2+1+2 |
| | **OR:** (i) Yes (1). (ii) Hormonal changes (1). (iii) Rest, talk, hobbies (1). (iv) If it persists, talk to a counsellor (2). | 1+1+1+2 |
| **39** | (i) Bob on a thread (1). (ii) Length (2). (iii) Clocks (2). | 1+2+2 |
| | **OR:** (i) A (1). (ii) 72 km and 54 km (2). (iii) A (1). (iv) 18 km (1). | 1+2+1+1 |
| **40** | (i) Gaps act as pinholes (1). (ii) Pinhole camera (1). (iii) Yes, bigger gap bigger spot (1). (iv) Straight-line travel (2). | 1+1+1+2 |
| | **OR:** (i) Mirror (1). (ii) Smooth surface (1). (iii) Regular and diffused (1). (iv) Still water, rough road (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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